



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/02

Paper 2

October 2025

FINAL MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **18** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme.
Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	excretory	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
1(b)	filters blood	1	Accept makes urine Accept removes urea from the blood Ignore cleans the blood or purifies blood or converts blood into urea or produces urea or filters urine Accept removes (excess) water from blood or controls water balance in the body
1(c)	bladder stores urine	2	each correct answer = 1 mark Accept releases urine or excreting urine Ignore contains urine or reference to urea or keeps urine Do not accept makes urine

Question	Answer	Marks	Further Information
2(a)	(idea that) mass does not change	1	Accept no change or stays the same or nothing
2(b)	(idea that) so no mass is 'lost'	1	Accept so no mass of the liquid is lost or so the mass does not change or because the mass will change (if liquid is lost) Accept (idea that) the mass will decrease (if liquid leaks out) Ignore references about accuracy or to make the reaction fair
2(c)	barium sulfate or BaSO ₄ sodium sulfate or Na ₂ SO ₄	2	each correct answer = 1 mark

Question	Answer					Marks	Further Information
3	television	(100 J electrical energy)	(60 J light energy)	(25 J sound energy)	15 J thermal energy	3	each correct answer = 1 mark Accept heat or heat energy for thermal energy Accept kinetic energy for movement energy Ignore missing units
	car engine	(100 J chemical energy)	(10 J movement energy)	40 J sound energy	(50 J thermal energy)		
	electric motor	(100 J electrical energy)	78 J movement energy	(12 J sound energy)	(10 J thermal energy)		

Question	Answer	Marks	Further Information
4	any four from M1 (idea of) dust cloud M2 (dust) obscured Sun or light (so less photosynthesis) M3 reduced photosynthesis M4 reduced supply of oxygen or amount of oxygen (in air) reduced M5 (idea that) plants died M6 (idea that) less food for herbivores so herbivores died or less feeding on plants M7 (idea that) less food for carnivores so carnivores died M8 more combustion M9 releases or more carbon dioxide in the air or releases or more greenhouse gases M10 climate change or global cooling M11 less habitat available	4	each correct answer = 1 mark Accept less producers Accept no food or less food if M6 and M7 not awarded Accept wildfires Accept causes large amount of carbon dioxide in the air Accept (idea of) global temperature decrease / ice-age Accept less shelter

Question	Answer	Marks	Further Information
5(a)(i)	C	1	Accept any indication of the correct answer, e.g. circling, ticking or underlining, but answer on answer line takes precedence
5(a)(ii)	30 (%)	1	Accept any value between 29 and 31
5(b)	any one from seas dried up rise in sea temperature or fall in sea temperature	1	Accept sea levels fall Accept possible ice-age or sea freezes Accept concentration of salts changed Accept underwater volcano Note increase in global temperature is not sufficient must refer to the temperature of the seas or oceans Ignore climate change or rise in sea level or flooding

Question	Answer	Marks	Further Information
6(a)	ions	1	Ignore positive and negative
6(b)	ionic positive and negative (ions) attract	2	Note covalent bonding = 0 marks for the question each correct answer = 1 mark Accept ions held together by electrostatic forces Accept attraction between oppositely charged particles Accept structure is giant ionic Note bonding can be used as an alternative to attraction.

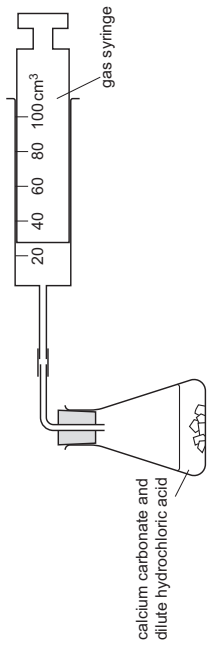
Question	Answer	Marks	Further Information
7(a)	(idea of) same thickness of (insulation) material	1	Accept (same) lid Accept (same) start temperature or (same) temperature of boiling water or (same) room temperature Note same temperature is not sufficient Accept (same) container or (same) material of container or (same) size of container Ignore volume of water or time
7(b)	thin copper (no mark) and (idea that) thin copper allows a lot of thermal energy transfer	1	Note mark is for explanation Note glass = 0 marks for the question Accept reverse argument e.g. glass is an insulator or glass allows little thermal energy transfer Accept copper is a better (thermal) conductor or copper is a good (thermal) conductor Note is a heat conductor is not sufficient
7(c)	risk of scalding (from boiling water)	1	Accept burning for scalding Note boiling water may splash is not sufficient

Question	Answer	Marks	Further Information
7(d)	D (idea that) D has smallest temperature drop	2	each correct answer = 1 mark Note if materials A, B, C or E given = 0 marks for the question Note if no material given accept the second marking point if correct Accept (idea that) has smallest temperature change Note smaller temperature change is not sufficient Accept D has the lowest temperature change = 2 marks Note the temperature decrease is 25 °C is not sufficient

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Question	Answer	Marks	Further Information
8(a)	convection currents (in mantle, magma or liquid rock) drag or push (tectonic plates)	2	each correct answer = 1 mark Accept hot magma rises and cool magma falls Accept convection (in mantle, magma or liquid rock) Note convection current makes tectonic plates move is not sufficient Accept tectonic plates float on mantle as an extra marking point
8(b)	earthquakes or volcanoes or tsunamis or mountains formed	1	Accept subduction Ignore one goes down and the other goes up

Question	Answer	Marks	Further Information
9(a)	minerals	1	more than one answer circled = 0 marks Accept any indication of correct answer, e.g. ticking or underlining, but circling takes precedence
9(b)(i)	photosynthesis	1	
9(b)(ii)	chloroplast	1	Ignore chlorophyll
9(c)	transpiration	1	Ignore evaporation or diffusion Accept evapotranspiration

Question	Answer	Marks	Further Information
10(a)	M1 appropriate method of gas collection e.g. gas syringe or collection over water using a graduated container M2 equipment must work to collect gas and be gas tight	2	M1 = 1 mark Note the equipment does not need to be part of a diagram Note gas syringe must either be labelled or clearly be a gas syringe, e.g. with graduations Note an unlabelled diagram with a graduated tube without the barrel is not sufficient as a gas syringe M2 = 1 mark Note the equipment must collect gas but not necessarily measure the volume Do not accept a single line for glass tube 

Question	Answer	Marks	Further Information
10(b)(i)	80 (cm ³)	1	
10(b)(ii)	yes (no mark) and (idea that the initial part of) line at 40 °C is steeper than line at 20 °C	1	Note if no = 0 marks Accept the reaction at 40 °C finishes before the one at 20 °C or at 40 °C finishes between 3.6 and 4.0 minutes and at 20 °C finishes between 6.0 and 6.4 minutes Accept correct comparison of the time to collect a known volume Accept at 40 °C it is faster to get to 80 cm³

Question	Answer	Marks	Further Information
11(a)	$\text{resistance} = \frac{\text{voltage}}{\text{current}}$ (resistance =) 96 Ω	3	96 Ω with or without working = 3 marks 96 with or without working = 2 marks correct equation or evidence of correct substitution into equation = 1 mark correct unit = 1 mark Accept $R = \frac{V}{I}$ Accept $R = \frac{24}{0.25}$ Accept ohms
11(b)	(current) decreases	1	Accept smaller Accept less Ignore weaker or slower current

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Question	Answer	Marks	Further Information															
12	<table><thead><tr><th>statement</th><th>true</th><th>false</th></tr></thead><tbody><tr><td>nebulae contain dust</td><td>☒</td><td>☐</td></tr><tr><td>nebulae only form moons</td><td>☐</td><td>☒</td></tr><tr><td>nebulae contain gases</td><td>☒</td><td>☐</td></tr><tr><td>nebulae are areas where stars form</td><td>☒</td><td>☐</td></tr></tbody></table>	statement	true	false	nebulae contain dust	☒	☐	nebulae only form moons	☐	☒	nebulae contain gases	☒	☐	nebulae are areas where stars form	☒	☐	2	all four correct = 2 marks two or three correct = 1 mark one correct = 0 marks more than one tick for a statement = 0 marks for that statement Accept any other indications of the correct answer, e.g. crosses, but ticking takes precedence
statement	true	false																
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Question	Answer	Marks	Further Information
13	any two from (idea of) increased sea levels (due to global warming) or flooding (increased sea level causes) animal to drown (idea of) reduced size of habitat (introduction of) a new predator (idea of) reduced food (idea of) new disease	2	each correct answer = 1 mark Accept environmental change for 1 mark Ignore island sinks into oceans Ignore animals cannot swim Accept not enough space on island or habitat destruction or no habitat or habitat loss Accept (idea of new) predation Ignore hunted by humans Accept no food (source) or increased competition for food Accept disease Ignore over population Ignore climate change or global warming

Question	Answer	Marks	Further Information
14(a)	decreases (from lithium to caesium)	1	Accept starts high and then becomes low
14(b)	(element) sodium (explanation) (idea that) trend is increasing from lithium to caesium or sodium is higher than potassium or (element) potassium (explanation) (idea that) trend is increasing from lithium to caesium or potassium is lower than sodium	2	correct element = 1 mark correct linked explanation = 1 mark
14(c)	any one from high or good thermal conductivity high or good electrical conductivity ductile malleable soft	1	Ignore reference to melting point and density Accept conducts heat Accept conducts electricity Accept can be cut easily with a knife Accept shiny when cut

Question	Answer	Marks	Further Information
15	When a small planet collided with the Earth the Moon was ejected during the collision.	1	more than one box ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence